

Design and Implementation of a Campus WiFi Bandwidth Sharing System with Reputation-Based Game Incentives

Zhitao WangJiaxuan WuXiangcheng Meng

Abstract— Aiming at the problems of uneven allocation of WiFi bandwidth resources and free-riding behavior among users in campus scenarios, a campus WiFi bandwidth sharing system based on a game theory incentive mechanism is designed and implemented. With FastAPI as the core development framework and combined with an SQLite database, a complete bandwidth sharing transaction platform is built, simultaneously establishing a credit system, a dynamic pricing mechanism and a coalition cooperation model oriented to game theory incentives. The system design incorporates key insights from recent advances in game-theoretic resource allocation: evolutionary game dynamics for strategy adaptation, cooperative game mechanisms for coalition formation with Shapley value benefit distribution, auction-theoretic approaches for dynamic resource pricing, and deep reinforcement learning for adaptive decision-making in heterogeneous network environments. By constructing multi-dimensional game perspectives involving players, strategies, and payoffs, the difficulty of promoting active sharing behavior is effectively solved, the phenomenon of free-riding is curbed, and a user strategy evolution path from non-cooperation to cooperation is realized.

Keywords: Campus WiFi; Bandwidth sharing; Game theory; Incentive mechanism; Dynamic pricing; Evolutionary game; Deep reinforcement learning

1. Introduction

With the popularization of the mobile Internet, campus WiFi has become a basic network facility for teachers and students in their daily study and life. However, there are prominent resource allocation problems in actual use: some users have surplus but idle bandwidth, while others lack bandwidth and their access experience is seriously affected. Free-riding behavior—consuming shared resources without contributing—is particularly common, leading to a gradual loss of willingness on the part of sharers, and overall system efficiency continues to decline, showing the dilemma of the "tragedy of the commons."

As a theoretical tool for studying strategic interactions among decision-making subjects, game theory provides an effective approach to solving the incentive and constraint problems in bandwidth sharing. In recent years, game-theoretic approaches have been successfully applied to a wide range of wireless communication problems: evolutionary game theory models strategy adaptation via replicator dynamics driving systems toward evolutionarily stable strategies (ESS); cooperative game theory with Shapley value enables fair resource allocation in spectrum sharing and coalition formation; auction theory provides market-based mechanisms for efficient dynamic resource

distribution; and deep reinforcement learning combined with evolutionary game dynamics has emerged as a powerful method for dynamic spectrum access in cognitive radio networks.

Based on the core ideas of game theory, this research designs and implements a campus WiFi bandwidth sharing system. By building a credit system, a dynamic pricing mechanism and a coalition cooperation mode, the system balances the "benefits of sharers" and the "costs of consumers", guides users to shift from free-riding to active sharing, and achieves the Pareto improvement of the whole system. The system adopts a front-end and back-end separation architecture, developed with FastAPI + SQLite + SSE, providing a user-friendly web interface for bandwidth sharing, consumption, credit management and coalition cooperation.

2. System Design

2.1. Design Objectives

The design objectives of the system are as follows:

- Establish a fair bandwidth sharing mechanism to motivate users to share surplus bandwidth.
- Curb free-riding behavior and set constraints on users who only consume but do not share bandwidth.
- Realize dynamic pricing to adapt to the real-time supply and demand changes of campus bandwidth resources.
- Support coalition cooperation and improve the overall resource utilization efficiency through group game theory.
- Ensure the usability and scalability of the system to adapt to changes in the scale of campus users.

2.2. Core Technical Architecture

The system adopts a front-end and back-end separation architecture. The back-end is developed based on the Python FastAPI framework, providing RESTful API interfaces. The data layer uses an SQLite database, and data operations are implemented through SQLAlchemy ORM. The front-end realizes real-time data push based on SSE (Server-Sent Events), displaying users' credit status and bandwidth pool dynamics. The core technology stack is: FastAPI (high-performance asynchronous web framework), SQLite + SQLAlchemy (data persistence), SSE (real-time communication), and Jinja2 (template engine).

2.3. Data Model Design

The core data models of the system encompass four interrelated entities that collectively cover the data storage needs of the entire bandwidth sharing process.

The **User** model serves as the foundational entity, storing core user information and behavioral metrics. Its key attributes include the username, a dynamically updated credit score reflecting reputation history, a virtual currency balance (initialized to 100 campus coins upon registration), cumulative 30-day shared and consumed bandwidth totals for sliding-window contribution tracking, and an optional coalition identifier linking the user to a cooperation group. These fields together enable the system to evaluate each user's standing within the incentive framework.

The **BandwidthShare** model records the bandwidth resources voluntarily contributed by users to the shared pool. Each record captures the sharer's identifier, the amount of bandwidth offered, an optional unit price set by the sharer, a peak-period flag indicating whether the contribution falls within high-demand intervals, and a validity period after which the share automatically expires. This model is the primary data source for the dynamic pricing engine and the bandwidth allocation matching algorithm.

The **BandwidthTransaction** model provides a complete audit trail of bandwidth consumption and sharing activities. Every transaction records the sharer's and consumer's identifiers, the exact bandwidth amount transferred, the final payment amount after applying all applicable discounts and penalties, and a transaction type field distinguishing between regular consumption, intra-coalition transfers, and penalty-flagged operations. These records enable retrospective analysis of system-wide resource flows and user behavior patterns.

The **Coalition** model stores group-level cooperation data, including the coalition name for identification, aggregated total shared bandwidth and total consumed bandwidth across all members, and a running total of saved discount amounts—the cumulative value of coalition-specific price reductions applied to intra-group transactions. This aggregate data is subsequently used for Shapley-value-based benefit redistribution among coalition members, ensuring that contributors receive proportional returns on their shared resources.

2.4. Design of Core Mechanisms Based on Game Theory

2.4.1. Credit Incentive and Constraint Mechanism

A multi-level credit system (Eagle, Tiger, Dolphin, Turtle) is designed, and users' credit scores are strongly correlated with their bandwidth sharing behavior. Users can get credit score rewards for sharing bandwidth, and the reward coefficient is positively correlated with users'

recent sharing rate and credit level. A small amount of credit scores will be deducted for consuming bandwidth. Turtle-level users are directly prohibited from consuming shared bandwidth. The user's contribution ratio is calculated based on a 30-day sliding window. Users with a contribution ratio lower than 10% will face a linearly increasing probability of consumption rejection, and long-term free-riding users will have their consumption eligibility permanently revoked.

This multi-tiered credit mechanism draws on principles from evolutionary game theory, where population dynamics drive agents toward evolutionarily stable strategies (ESS). The credit system steers the campus user population toward a cooperative equilibrium through carefully calibrated rewards and penalties.

2.4.2. Dynamic Pricing Mechanism

The basic bandwidth price is dynamically adjusted with the supply and demand changes of the global available bandwidth pool. Taking 500 MB as the benchmark, the price will increase by 10% when the available bandwidth is less than 400 MB, and decrease by 10% when it is more than 600 MB. Multi-level discounts are set according to users' credit level, coalition identity and ranking list position: 30% discount for Eagle users, 8% discount for intra-coalition users, and an additional 5% discount for the top 10% users on the ranking list. A price penalty is imposed on free-riding users: the consumption price is doubled for users with a contribution ratio lower than 10%, and increased by 50% for those with a ratio between 10% and 20%.

The dynamic pricing approach is inspired by auction-based resource allocation mechanisms demonstrated in heterogeneous networks, where users submit bids reflecting their valuation of network resources and the auctioneer determines efficient allocations through market-clearing prices.

2.4.3. Coalition Cooperation and Shapley Value Allocation

Users are supported to create or join bandwidth coalitions. Intra-coalition users are given priority in bandwidth consumption matching and enjoy additional discounts. The "total saved discount amount" within the coalition is allocated according to the proportion of bandwidth shared by members based on the Shapley value allocation principle, ensuring fairness in benefit distribution within the coalition and encouraging members to increase their contribution.

The coalition cooperation mechanism embodies the cooperative game paradigm that has proven effective in wireless resource sharing, as evidenced by cooperative game-aided spectrum sharing in cognitive radio networks.

2.4.4. Guarantor Mechanism

Tiger- and Eagle-level users can act as guarantors to provide credit guarantees for new users: Guaranteed users get an increase in credit limit and an initial reward of campus coins. Guarantors can obtain a continuous commission of 1% of the bandwidth consumption amount of guaranteed users, forming a positive incentive for high-credit users to actively invite and assist newcomers.

3. System Implementation

3.1. Implementation of Core Function Modules

3.1.1. User Module

The user module realizes the functions of user registration, login and information query. User credit scores (initialized to 100 points) and virtual currency (initialized to 100 campus coins) are set during registration. User online status and last active time are updated during login. The interface /user/profile returns the user's credit score, campus coin balance and 30-day bandwidth sharing data.

3.1.2. Bandwidth Sharing and Consumption Module

For bandwidth sharing, after a user submits the amount of shared bandwidth, the system records the bandwidth resource information, calculates and rewards credit scores, and updates the 30-day sharing data. For bandwidth consumption, the interface /api/cost-preview first previews the consumption cost, credit changes and rejection probability, and then executes the allocation logic—prioritizing intra-coalition bandwidth resource matching, verifying the user's campus coins and credit level, deducting fees, and updating consumption records.

3.1.3. Coalition Module

The coalition module supports coalition creation, joining and information query. The basic coalition information is initialized when a coalition is created, and the coalition member association is updated when a user joins the coalition.

3.1.4. Real-time Push Module

The real-time push module realizes the real-time push of campus coin and bandwidth pool data based on SSE. After a user consumes or shares bandwidth, the system pushes campus coin changes to relevant users and sends bandwidth pool update notifications to all online users.

3.2. Examples of Core Code Implementation

3.2.1. Credit Level Calculation

Listing 1. Reputation level judgment function.

```
REPUTATION_LEVELS = [
    ("diamond", 90), ("gold", 70),
    ("silver", 50), ("bronze", 0),
]
```

```
def get_reputation_level(score):
```

```
    for level, threshold in REPUTATION_LEVELS:
        if score >= threshold:
            return level
    return "bronze"
```

3.2.2. Dynamic Base Price Calculation

Listing 2. Dynamic base price calculation function.

```
def get_dynamic_base_price(db):
    shares = db.query(BandwidthShare)
    .filter(BandwidthShare.is_available).all()
    total = sum(s.bandwidth_amount for s in shares)
    if total <= 0: return 0.12
    ratio = total / 500.0
    if ratio < 0.8: return 0.12 * 1.1
    if ratio > 1.2: return 0.12 * 0.9
    return 0.12
```

3.2.3. Core Logic of Bandwidth Allocation

Listing 3. Bandwidth allocation manager (core interface).

```
class BandwidthManager:
    @staticmethod
    def allocate(user_id, amount, db):
        # Match coalition resources first
        # Verify coins & credit level
        # Deduct fees & update records
        return result, error
```

3.3. System Interface Demonstration

Figures 1–3 show the main interfaces of the running system, demonstrating the login/registration page, bandwidth sharing and consumption dashboard, and the user home page with credit and coin overview.

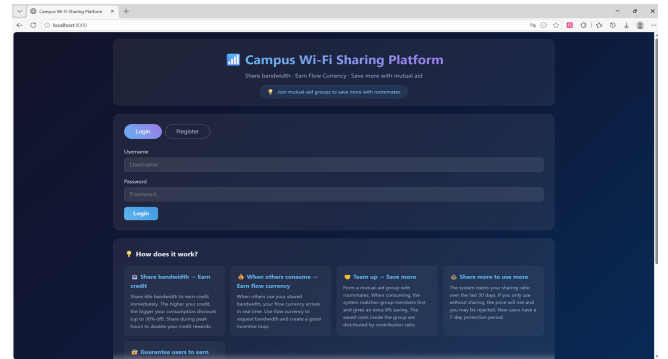


Fig. 1. Login and registration page with feature overview cards.

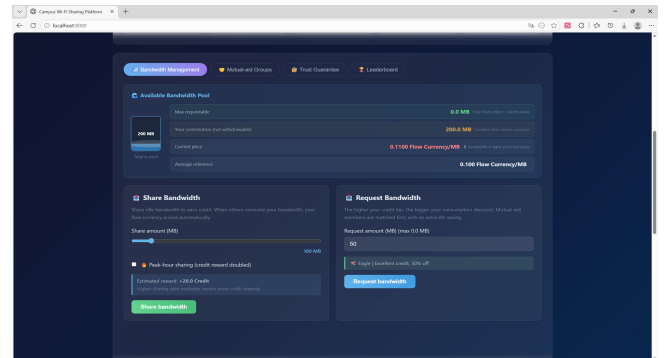


Fig. 2. Bandwidth sharing and consumption interface with dynamic pricing preview.

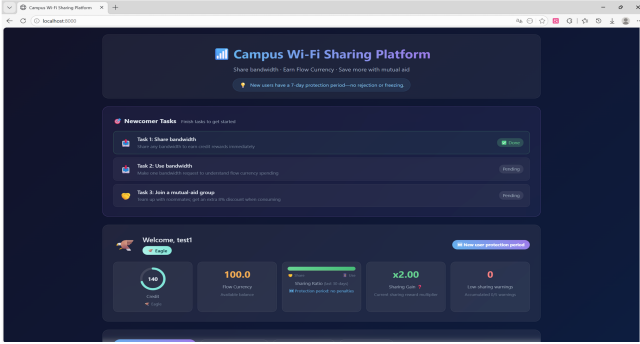


Fig. 3. User dashboard showing credit score, flow currency, sharing ratio, and newcomer tasks.

4. System Testing and Analysis

4.1. Functional Testing

Full coverage testing was carried out on all core function modules. The **user registration and login** module was verified to function normally, with correct initialization of credit scores (100 points) and virtual currency (100 campus coins) upon account creation, and proper updating of online status and last-active timestamps during login. The **bandwidth sharing** module correctly updated credit scores and sharing records upon submission; the reward coefficient scaled appropriately with the user's credit level and recent sharing rate. **Bandwidth consumption** testing confirmed that price calculation, multi-level discount application (credit-based, coalition-based, and ranking-based), penalty surcharges for free-riders, and campus coin deduction were all accurate across diverse user profiles. The **coalition creation and joining** workflow was verified: coalition information (name, aggregate shared/consumed bandwidth, saved discount totals) was correctly initialized upon creation, and member association records were properly updated upon joining. Finally, **real-time push** functionality was confirmed: campus coin balance changes were pushed to affected users, and bandwidth pool updates were broadcast to all online users within acceptable latency bounds via the SSE channel.

In summary, every tested module met its functional specification. Core game-theoretic mechanisms—credit-driven access control, dynamic price adjustment, and coalition-internal benefit redistribution—operated as designed, and the system demonstrated robust behavior under edge cases including zero-balance users, free-riders with low contribution ratios, and concurrent coalition operations.

4.2. Effectiveness Analysis of the Incentive Mechanism

The incentive mechanism of this system is designed around the three core elements of game theory: players, strategies and payoffs. By reconstructing the game rules

for campus WiFi bandwidth sharing, it breaks the prisoner's dilemma in the traditional non-incentive mode and realizes the unification of individual rationality and collective rationality.

All campus WiFi users are defined as the game players, and each player has two core strategic options: active bandwidth sharing and free-riding (consuming bandwidth only without sharing). The payoff structure corresponding to different strategies is set in a differentiated manner by several sub-mechanisms.

In the bandwidth sharing scenario without an incentive mechanism, free-riding becomes the dominant strategy for players: participants can consume the bandwidth shared by others free of charge without any resource cost, and the short-term zero-cost payoff is much higher than that of active sharing. However, when all players adopt this strategy, the total amount of bandwidth resources shared system-wide tends toward zero, and all players' long-term payoffs decrease, leading to the classic collective action problem.

This system reconstructs the game payoff structure precisely, making active bandwidth sharing the rational optimal strategy for users. Multi-level positive incentives are designed for active sharing behavior, where participants can obtain practical benefits such as reputation score improvement, virtual currency rewards and consumption discounts. Meanwhile, a negative constraint mechanism is imposed on free-riding behavior, and users' continuation of free-riding will face cost increases and access restrictions. Under this rule system, users' strategic choices gradually tilt toward active sharing, and the game relationship of bandwidth sharing shifts from non-cooperation to cooperation, ultimately reaching a win-win Nash equilibrium.

This payoff restructuring is consistent with the evolutionary game dynamics studied in recent wireless communication research. When agents in a population repeatedly interact under well-designed payoff functions, replicator dynamics drive the system toward an evolutionarily stable strategy where cooperation becomes the dominant behavior.

5. Conclusion and Prospect

Based on game theory, this research constructs a campus WiFi bandwidth sharing system. Through mechanisms such as credit incentives, dynamic pricing and coalition cooperation, the system effectively solves the problems of idle campus bandwidth resources and free-riding, and improves resource utilization efficiency. The system is developed based on the FastAPI framework, featuring a clear technology stack, complete functions and good usability.

The further optimization directions in the future are as follows:

1. Introduce machine learning models, particularly deep reinforcement learning approaches as demonstrated in dynamic spectrum access research, to accurately predict the supply and demand changes of bandwidth and optimize the dynamic pricing strategy.
2. Support multi-terminal adaptation and develop a mobile app to improve the convenience of user operation.
3. Combine blockchain technology to realize the immutability of bandwidth transaction records and further ensure the fairness of sharing.
4. Extend the evolutionary game framework to incorporate more sophisticated strategy spaces, where agents adapt not only their binary sharing decisions but also continuous parameters such as sharing quantity and timing.
5. Integrate auction-theoretic refinements such as combinatorial auctions and Vickrey-Clarke-Groves (VCG) mechanisms to handle more complex resource allocation scenarios involving multiple bandwidth tiers and quality-of-service requirements.

References

- [1] "Reflection Coefficients Optimization in IRS-assisted Communication: An Evolutionary Game Approach," *2021 13th Int. Conf. on Wireless Communications and Signal Processing (WCSP)*, Changsha, China, 2021, pp. 1–5.
- [2] "Dynamic Spectrum Access in Cognitive Radio Networks Using Deep Reinforcement Learning and Evolutionary Game," *2018 IEEE/CIC ICC*, Beijing, China, pp. 405–409, Aug. 2018.
- [3] "Auction-Based Users Dynamic Allocation in Heterogeneous Networks," *2018 IEEE/CIC ICC*, Beijing, China, pp. 636–640, Aug. 2018.